Myungjin Lee 이명진

M.S.E. in Computer Science & Artificial Intelligence • Dongguk University

AI for Drug Discovery • ADMET Prediction • Drug Repurposing • Graph Learning • XAI

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Research Interests

AI for Drug Discovery: Molecular Graph Representation Learning, ADMET Property Prediction, Drug Repurposing, Drug–Target Interaction (DTI), Drug Response Prediction, Explainable AI (XAI).

ML & Data Mining for Bioinformatics: Multi-omics Data Integration, Network Biology, Differential Expression Analysis, Dimensionality Reduction, Multimodal AI for Biomedical Problems.

Education

Mar 2024 – Aug 2026
M.S.E. in Computer Science and Artificial Intelligence
Dongguk University, Seoul, Korea
Major: AI in Healthcare and Medicine • GPA 4.04/4.5

Mar 2023 – Feb 2024
B.E. in Industrial & Management Engineering (Advanced Major Program)
Induk University, Seoul, Korea
Convergence Minor in Deep Learning • GPA 4.43/4.5 • Valedictorian

Mar 2020 – Feb 2023
A.E. in Industrial & Management Engineering
Induk University, Seoul, Korea
GPA 4.41/4.5 (Major 4.42/4.5) • Valedictorian

Research Experience

Mar 2024 – Present
PRISM Lab, Dongguk University, Seoul, Korea
Position: Graduate Researcher (M.S.E.) — Advisor: Dr. Sangsoo Lim
Role: Molecular graph representation learning and explainable AI for drug discovery — interpretable ADMET prediction from molecular substructures (first-author SJoINT; co-author MAGNET); a drug-repurposing model developed and registered to the national bio-data platform (KBDI); and bulk / single-cell RNA-seq omics analysis for target discovery.

Manuscripts

* Corresponding author; underlined name is the CV owner.

2026
Lee M., Mo J., Kang M., Lim S.*
SJoINT: Substructure-Driven Junction Tree for Interpretable ADMET Prediction
In preparation
Dual-view learning of atom-level graphs and substructure-level junction trees via structure-constrained bidirectional cross-attention with augmentation-free contrastive pretraining (ZINC250K); evaluated on MoleculeNet ADMET and MoleculeACE activity-cliff tasks.

2026
Mo J., **Lee M.**, Lee S., Lim S.*
MAGNET: Cross-view Molecular Graph Learning for Interpretable ADMET Prediction
Under review · *Bioinformatics*
Multi-view molecular graph framework integrating BRICS, junction-tree, and Murcko fragmentations via cross-view meta-graphs, with self-supervised pretraining on ZINC250K and ChemBERTa-enhanced fine-tuning across 10 MoleculeNet ADMET benchmarks.

Conference Presentations

Jun 2026
Structure-Constrained Bidirectional Cross-Attention for Self-Supervised Molecular Representation Learning
Poster, Korea Computer Congress (KCC) 2026, KIISE, Korea.

Oct 2025
Weighted Junction-Tree Nodes for Enhanced Interpretability in ADMET Tasks
Poster, BIOINFO 2025, KSBI, Korea.

Grants

2024 – 2025
Master's Student Research Grant (석사과정생 연구장려금지원사업) --- **Principal Investigator**
Ministry of Science and ICT (MSIT). Individual research grant on functional-group junction-tree-based multi-task molecular graph representation learning. Advisor: Dr. Sangsoo Lim.

Research Projects (Participating Researcher)

2025 – 2026 **Multimodal AI-based Target Discovery and Drug Validation for Aging-related MASLD**
Individual Basic Research Program, NRF (MSIT). PI: Dr. Sangsoo Lim; Collaborator: Dr. Gung Lee (Mayo Clinic, USA).
Preliminary omics/bioinformatics for target discovery: curated public & experimental bulk/single-cell RNA-seq; QC → DEG → GSEA/ssGSEA → module scoring; characterized lipid-metabolism reprogramming and cellular-senescence signatures (SenMayo/SASP) in aging mouse liver, with a supporting cross-check in public human data; prioritized candidate genes and Plin2-defined senescent-cell subsets.

2025 – 2026 **Open AI Drug Discovery and Data Analysis Platform Center (Bio-SYNAPSE)**
Bio & Medical Technology Development Program, NRF (MSIT). PI: Dr. Minho Lee (host).
Drug repurposing: built a standardized PrimeKG-based benchmark to evaluate five models (TxGNN, DREAMWalk, DRHGCN, AMDGT, SVGA) across 190 diseases / 3,379 drugs; developed DC-VGAE and a performance-based ensemble. **ADMET:** benchmarked and developed an interpretable ADMET model using junction-tree / Murcko / BRICS substructures with cross-attention. **Deployment:** packaged the drug-repurposing model as a Docker image and registered it to the national bio-data platform (KBDI).

Competitions

Aug 2024 **2024 Samsung AI Challenge — Machine Learning Force Fields** Top 10% (11th)
Samsung Electronics SAIT · DACON. Built MLFF models predicting atomic energies and forces with uncertainty quantification for semiconductor-material simulation (team DAI).

2024 **DREAM Olfactory Mixtures Prediction Challenge**
International DREAM Challenge, Sage Bionetworks. Graph neural network predicting olfactory similarity of molecular mixtures.

Honors & Awards

2023 **Highest Academic Achievement Award**
Induk University, Korea.

2020 – 2023 **Merit-based Academic Scholarship** — awarded every semester
Induk University, Korea (incl. GRAPE Talent Scholarship).

Patents & Software Registrations

Oct 2025 An Embedding Method via Multi-view Graph Integration of Molecular Structures
Software Registration, Korea Copyright Commission, No. C-2025-040207.

Jan 2025 System and Method for Predicting Olfactory Characteristics of Odor Mixture Data
Korean Patent Application (pending), No. 10-2025-0003146.

Nov 2024 Deep Learning Model for Similarity Prediction of Graph-based Mixtures
Software Registration, Korea Copyright Commission, No. C-2024-042597.

Technical Skills

Programming	Python, R (basic), Java, C#, JavaScript, SQL, HTML
Deep Learning	PyTorch, PyTorch Geometric, Hugging Face Transformers, scikit-learn
Data Analysis & Visualization	pandas, NumPy, anndata, seaborn, matplotlib
Cheminformatics	RDKit, DeepChem, TorchDrug, ChemPy
Bioinformatics	Scanpy, Seurat, bulk & single-cell RNA-seq, differential expression (DESeq2 / PyDESeq2, MAST), GSEA / ssGSEA / GSVA, GO enrichment
Tools & Databases	Git, Linux, Docker, MySQL

Languages

Korean	Native
English	TOEIC 860 (LC 440 / RC 420)